

How Our Foresight Pipeline Works

Required reading *before* you build slides

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Abstract

This document explains, **from scratch and assuming no prior knowledge**, what our pipeline does, why it is built this way, and – most importantly – **what our actual result is**. Read it fully before making slides. At the end you will find a glossary and a list of “sentences you must never say”. If you only have five minutes, read Section 1 (the big picture) and Section 7 (the central finding).

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1 What is this even about?

We are **not** building a tool that “predicts the future”. We are building a **foresight pipeline**: an automated machine that turns a collection of documents (“what do we know about this field?”)

into a **map of plausible futures** – structured, traceable, and without a human having to invent the scenarios by hand.

Two things are central and should appear in every presentation:

- **Domain-agnostic.** The actual product is a *generic* framework. You dock any knowledge base onto it (spectrum monitoring, agriculture, whatever) and the pipeline derives the rest by itself. *Spectrum monitoring is only our test case* (the sponsor, Rohde & Schwarz), not the purpose.
- **Traceability.** Every statement, every driver, every scenario carries references (`source_chunk_ids`) back to the concrete passages in the knowledge base. Nothing is “hallucinated” without a source you can look up.

The one-sentence explanation

The pipeline reads a knowledge base, distills the *driving uncertainties* of a field, spans a space of plausible scenarios from them, and evaluates that space honestly – including the question of whether clear “archetypes” exist at all, or only a continuum.

2 The mental model (big picture)

Before the individual steps, the picture to hold in your head. Everything splits into four phases:

1. **Upstream (what is the world made of?)** – *drivers* are extracted from documents and their possible states are defined.
2. **Scenario generation (which combinations are coherent?)** – the drivers are coupled (which one promotes/inhibits which?) and coherent combinations = scenarios are produced. We do this with *two* methods in parallel.
3. **Analysis (what structure does the scenario space have?)** – are there clearly separable groups (*archetypes*) or is everything a gradual transition (*continuum*)? This is where our main finding lives.
4. **Evaluation & synthesis (what does it mean strategically?)** – scenarios are scored (multi-criteria), placed in time, and summarized strategically.

The single most important concept: driving vs. response

We separate **driving axes** (exogenous uncertainties of the *environment* – regulation, market, geopolitics, technology push) from **response axes** (the actor’s own product/technology, e.g. hardware/software). **Scenarios are spanned by the driving axes**; the technology is what *responds* and gets evaluated per scenario. Whoever mixes these two up explains the whole method wrong.

3 The pipeline step by step

3.1 Knowledge Base (KB)

All source PDFs and arXiv papers are split into small text *chunks*, turned into vectors (*embeddings*) and stored in a vector database (Chroma). Currently $\approx 3,900$ chunks for the spectrum test case. Each chunk has a `pool`: `product` (describes our product) or `trend` (describes the environment). This is the only docking point – everything else is derived.

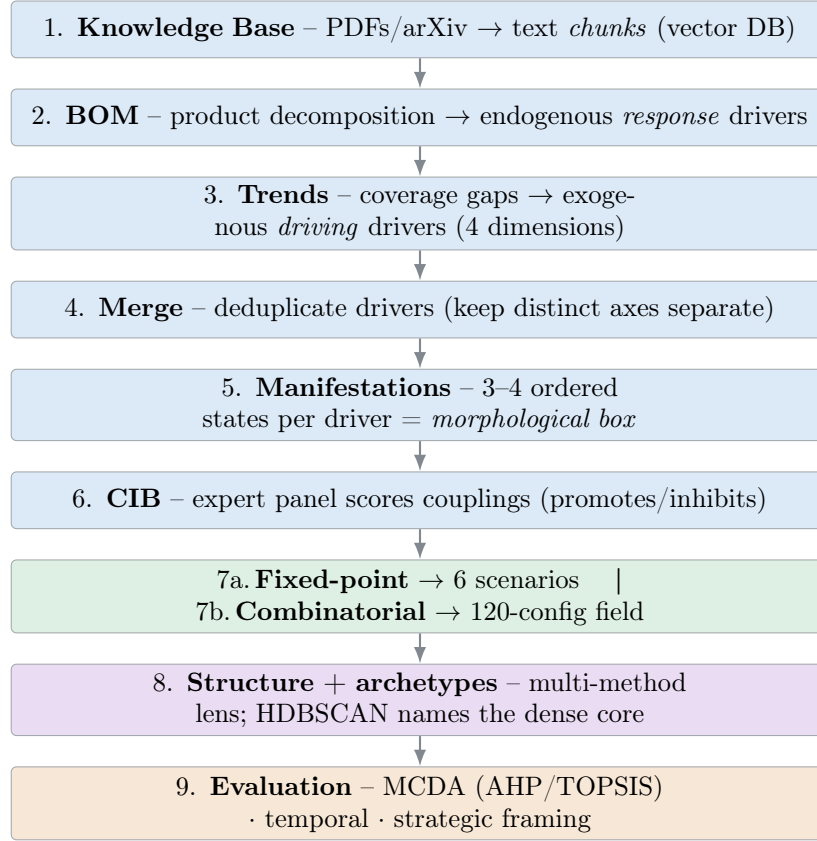


Figure 1: The data flow. Blue = upstream, green = scenario methods, purple = analysis, orange = evaluation.

3.2 BOM – the product decomposition

BOM = *Bill of Materials*. The R&S product is decomposed into its building blocks (ADC, digital downconverter, direction-finding modules, ...). These become **response drivers** – they are the *system* under study, not the driving uncertainty.

3.3 Trends – driving uncertainties from the coverage gaps

Here is an elegant idea: a trend chunk is interesting exactly when *none* of the BOM building blocks already covers it (similarity below a threshold). Such chunks are called **orphans** – they describe environment the product does not yet capture, i.e. potential *driving* uncertainties. Flow:

1. Each orphan is assigned to its nearest **driving dimension**: *regulatory, market, geopolitical, technological* (generic, domain-neutral anchors).
2. Within each dimension we cluster; granularity scales with how much content that dimension has (a rich dimension → more, more distinct drivers).
3. An LLM formulates one driver per cluster and **stamps** its dimension onto it.

Source cap: no single document may contribute more than N chunks (currently 150) to the orphan pool. Without it, two huge regulatory documents (WRC-23, ITU Handbook) would drown everything else.

3.4 Merge – consolidate, but keep the axes apart

BOM and trend drivers are deduplicated (same names, similar embeddings, LLM grouping). New and important: a **cross-dimension guard** prevents drivers of *different* dimensions from being merged together. Afterwards each driver has an `axis_role` (driving/response). For the expensive CIB step the **top 14** drivers are selected (weighted by origin + amount of evidence).

3.5 Manifestations – the morphological box

For each driver an LLM generates 3–4 **states** (manifestations), *ordered* from optimistic to pessimistic. Drivers $\times$ states = the **morphological box** (*Zwicky box*). A *scenario* is simply: pick exactly one state from each driver.

3.6 CIB – Cross-Impact Balance (the heart)

CIB (after Weimer-Jehle) asks, for *every pair of drivers*: “If driver A takes this state – does that promote (+) or inhibit (-) a particular state of driver B?” Instead of a single opinion we use a **panel of several AI personas** + two *Delphi* rounds (the personas see the outliers and may revise). Two specifics:

- **Dissent-preserving:** if a minority sees a strong trade-off (score ≤ -2), it is *not* averaged away by the median. Reason: LLMs are too positive; without this we would get almost only “everything promotes everything” and the scenario space collapses.
- **Calibration:** we measure the share of negative entries in the matrix. $\approx 24\%$ negative sits in the literature-normal range (20–30%). Before, it was near 0% – a known LLM positivity bias.

3.7 Two routes to scenarios (deliberately parallel)

From the box + CIB matrix, scenarios are produced in **two** ways – both kept as equal-standing views:

- **7a. Fixed-point method (classic Weimer-Jehle).** Find *self-consistent equilibria*: configurations where no single driver “wants to flip”. These are the **fixed points** (attractors). From them, 6 maximally different, typed scenarios are chosen.
- **7b. Combinatorial method.** Instead of the hard fixed-point gate, a *soft* filter (a “contradiction ratio”): sample ≈ 120 plausible configurations. This yields a *whole field* instead of a few points – the basis for the archetype analysis.

For each configuration an LLM writes a **grounded narrative** (with RAG evidence from the KB).

3.8 Landscape – the map

The scenario space is visualized (UMAP for looks, PCA for *interpretable* axes) and annotated with an honest structure verdict (see Section 5).

3.9 Evaluation, time, strategy

- **Grounded evaluation / MCDA:** an evidence-based auditor scores each scenario along several criteria (impact, probability, actionability, time horizon, risk), *with evidence* per score. MCDA (AHP weights + TOPSIS ranking) turns this into a ranked list.
- **Temporal / DVI:** places drivers by signal maturity (emerging vs. established).
- **Strategic framing:** an LLM synthesizes the whole analysis into strategic statements.

4 Why TWO scenario methods?

They answer different questions and share the same upstream (KB → drivers → CIB) and the same downstream (narratives → landscape → evaluation). Only the selection step differs.

	Fixed-point (7a)	Combinatorial (7b)
Question	Where does the system <i>rest</i> ? (equilibria)	What is plausible at all? (whole field)
Filter	hard (true fixed points)	soft (contradiction ratio)
Output	few (6) typed scenarios	large field (≈ 120)
Strength	classic, established, comparable	basis for clusters/archetypes

Framing for the slides

“We offer the classic Weimer-Jehle fixed-point analysis *and* a field-based combinatorial analysis – both from the same, computed-once upstream.” That is credible and honest.

5 Structure: are there archetypes or a continuum?

The dream of every scenario study: “here are the 3–4 archetype scenarios.” But you may only claim that if the space *actually* splits into separable groups. So we measure it instead of asserting it.

Silhouette & null model. The *silhouette* measures (0–1) how cleanly separated clusters are. Rule of thumb: below **0.25** = no usable clusters. We additionally compare against a **uniform-random null model** (same field, but randomly filled): is our structure significantly larger than pure chance?

Multi-method lens. The crux: the standard method (KMeans) *forces every point into a cluster* and thereby misses structure. So we look through several “lenses”:

- Encoding: *one-hot* vs. *ordinal* (ordinal keeps the optimistic→pessimistic order).
- Method: *KMeans* (all points) vs. *HDBSCAN* (density-based, allowed to leave points as “noise”, unclustered).

Do not compare apples with oranges

The HDBSCAN silhouette applies only to the *dense part* (noise excluded) and is **not** directly comparable to the KMeans silhouette (all points). On slides, always put them side by side – do not sell it as “we improved the silhouette from 0.07 to 0.4”.

6 The central finding (this must stick)

After all improvements (more & more independent drivers, a de-skewed corpus, a de-biased CIB), the honest, measured result is:

The field is a HYBRID

It is **neither** a cleanly clustered space **nor** pure chaos, but a **dense archetypal core** (a minority of scenarios form real, well-separated groups) **embedded in a continuum** (the majority blend gradually into one another). Standard KMeans hid the core; **HDBSCAN + ordinal encoding** reveal it.

Concretely (spectrum test case, combinatorial field, 120 configurations):

- KMeans over all points: silhouette ≈ 0.07 (looks like “no structure”).

- HDBSCAN + ordinal: finds **7 dense clusters** (silhouette ≈ 0.39); $\approx 37\%$ of points remain as a *continuum halo*, unclustered.
- These 7 clusters are **named** by an LLM (e.g. “Assured Spectrum Commons”, “Contested Sharing Governance”, “Algorithmic Spectrum Stewardship”, ...), each with the driver states that distinguish it from the others.

Why is the engine nonetheless correct (and why that is our selling point)? We have a *positive control*: an artificial field with *genuine* block structure yields silhouette ≈ 0.72 – the machine *does* find clear clusters when they exist. Our real field looks different because the domain is that way, not because the tool fails. **The engine measures the input property honestly and does not hallucinate clusters.** That integrity is exactly the point.

7 Honest limitations (state them proactively)

- **Continuum halo:** the archetypes describe the *core*, not the whole space. Always quote the halo’s percentage.
- **Corpus skew:** the geopolitical dimension is still thin (1–2 drivers). Enriching it with more / opposing sources is on the roadmap.
- **MCDA compression:** the evaluation LLM often gives very similar scores (impact/probability barely spread). That is a *separate* lever (score anchoring), not part of the structure finding.
- **Variance:** results are seeded but LLM-dependent – quote numbers as *ranges*.

8 Run it yourself

- Whole pipeline (one command): `CIB_DISSENT_PRESERVING=1 uv run python run_full.py -skip-arxiv`
- Dashboard: `uv run uvicorn web.app:app -port 8000` → `http://localhost:8000` (pages *Archetypes* and *Landscape* → Combinatorial → “Structure — method comparison”).
- Tests: `uv run pytest -q`. **Always uv, never pip/venv.**

Sentences you must never say

Forbidden sentences (and the correct version)

- “The tool predicts the future.” → It *structures plausible scenarios*.
- “We found 7 clean archetypes that cover everything.” → 7 archetypes in the *dense core*; the rest is honestly a continuum.
- “The low silhouette means it didn’t work.” → It is an honest measurement; the hybrid finding *is* the result.
- “The technology drivers are our scenario axes.” → No: driving (environment) spans the space, response (technology) reacts.
- “We raised the silhouette from 0.07 to 0.4.” → Different lenses, not comparable.
- “It only works for spectrum.” → Domain-agnostic; spectrum is the test case.

Glossary

Knowledge Base / chunk

The docked document collection; a chunk is a small text excerpt with a vector embedding in the vector DB.

BOM

Bill of Materials – the decomposition of the product into building blocks (= response drivers).

Driver

A factor shaping the future. **Driving** = exogenous uncertainty (environment), **response** = endogenous product/technology capability.

Dimension

Category of a driving driver: regulatory, market, geopolitical, technological.

Orphan chunk

A trend chunk not covered by any BOM block – the source of driving drivers.

Morphological box (Zwicky box)

Drivers $\times$ ordered states; a scenario = one state per driver.

CIB

Cross-Impact Balance – pairwise coupling matrix (promotes/inhibits), elicited by a persona panel + Delphi.

Fixed point / attractor

A self-consistent configuration; no driver wants to flip on its own.

Contradiction ratio

A soft measure (0–1) of how strongly a configuration violates the CIB couplings.

Silhouette

A measure (0–1) of cluster separation; < 0.25 = no usable clusters.

Null model

Random baseline: is the structure larger than pure chance (z-score)?

Effective dimension

The “effective number of axes”; high = isotropic/spread out, no dominant axis.

KMeans / HDBSCAN

Clustering methods: KMeans forces every point into a cluster; HDBSCAN is density-based and allows noise.

Ordinal encoding

Driver state as a number 0 (optimistic) ... 1 (pessimistic); keeps the order that one-hot loses.

Archetype

A named, dense cluster of scenarios.

Continuum halo

The gradual remainder not assigned to any archetype.

MCDA / AHP / TOPSIS

Multi-criteria evaluation: AHP weights the criteria, TOPSIS builds the ranking.

Traceability

Every statement is traceable back to source chunks.

Dissent-preserving

A CIB aggregation that does not average away strong minority dissent.